HOSPITAL BILLING AND PATIENT CLASSIFICATION

Python Project Report

Project Overview

This project calculates hospital bills by combining consultation, diagnostic, and treatment charges. It applies a conditional concession, validates billing rules, and classifies patients according to their payment status.

Policy used for this example

Because no exact payment policy was included in the topic, this report uses an example policy: bills with a subtotal of ■5,000 or more receive a 10% concession. Patients are classified as Paid in Full, Partially Paid, or Unpaid based on the amount paid and the remaining balance.

Prepared as a simple, beginner-friendly Python project.

1. Problem Statement

Develop a Python program to process hospital consultation, diagnostic, and treatment charges; calculate the total bill; apply conditional concessions; validate billing constraints; and classify each patient according to a payment policy.

2. Objectives

• Calculate the subtotal from consultation, diagnostic, and treatment charges.

• Apply a concession when the bill meets the policy condition.

• Calculate the net bill, amount paid, and remaining balance.

• Validate that charges and payments are not negative.

• Prevent payment from exceeding the net bill.

• Classify patients as Paid in Full, Partially Paid, or Unpaid.

3. Assumptions and Billing Policy

• All amounts are entered in Indian rupees (■).

• Consultation, diagnostic, treatment, and paid amounts must be zero or positive.

• A 10% concession is applied if the subtotal is ■5,000 or more; otherwise, no concession is applied.

• Payment cannot be greater than the net bill.

• Paid in Full: balance is zero; Partially Paid: some amount is paid but a balance remains; Unpaid: no amount has been paid.

4. Algorithm

1. Start the program.

2. Enter patient details and charges.

3. Validate that all charges and payment are non-negative.

4. Calculate subtotal = consultation + diagnostic + treatment.

5. If subtotal is ■5,000 or more, calculate concession = 10% of subtotal; otherwise concession = 0.

6. Calculate net bill = subtotal – concession.

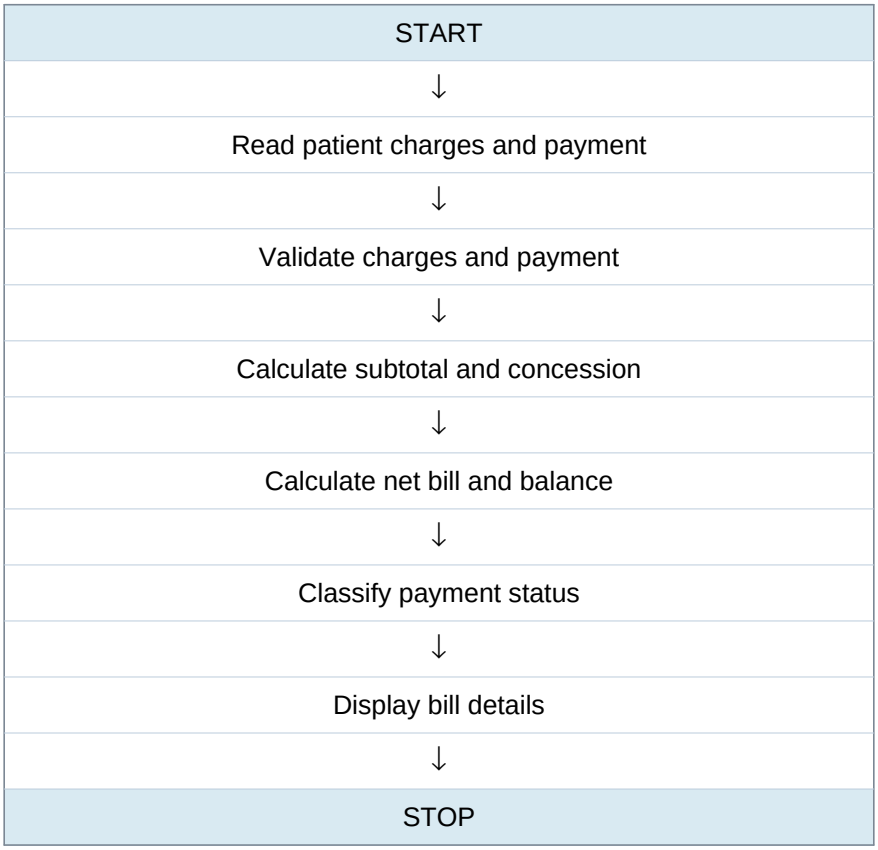
7. Check that paid amount does not exceed the net bill.

8. Calculate balance = net bill – paid amount.

9. Classify the patient using the balance and paid amount.

10. Display bill details and stop.

5. Flowchart



6. Pseudocode

```
BEGIN
    INPUT patient name, consultation, diagnostic, treatment, paid
    IF any amount is negative:
        DISPLAY error
    ELSE:
        subtotal = consultation + diagnostic + treatment
        IF subtotal >= 5000:
            concession = subtotal * 10 / 100
        ELSE:
            concession = 0
        net_bill = subtotal - concession
        IF paid > net_bill:
            DISPLAY payment error
        ELSE:
            balance = net_bill - paid
            IF balance == 0: status = "Paid in Full"
            ELSE IF paid > 0: status = "Partially Paid"
            ELSE: status = "Unpaid"
            DISPLAY subtotal, concession, net_bill, paid, balance, status
END
```

7. Python Source Code

Save the following file as **hospital_billing_classification.py** and run it in VS Code.

```
# Hospital Billing and Patient Classification

def calculate_bill(patient):
    consultation = patient["consultation"]
    diagnostic = patient["diagnostic"]
    treatment = patient["treatment"]
    paid = patient["paid"]

    # Validate billing constraints
    charges = [consultation, diagnostic, treatment, paid]
    if any(amount < 0 for amount in charges):
        raise ValueError("Charges and payment cannot be negative.")

    subtotal = consultation + diagnostic + treatment

    # Assumed policy: bills of Rs. 5000 or more receive a 10% concession.
    if subtotal >= 5000:
        concession = subtotal * 0.10
    else:
        concession = 0

    net_bill = subtotal - concession

    if paid > net_bill:
        raise ValueError("Payment cannot be greater than the net bill.")

    balance = net_bill - paid

    if balance == 0:
        status = "Paid in Full"
    elif paid > 0:
        status = "Partially Paid"
    else:
        status = "Unpaid"

    return {
        "name": patient["name"],
        "subtotal": subtotal,
        "concession": concession,
        "net_bill": net_bill,
        "paid": paid,
        "balance": balance,
        "status": status
    }


def main():
    patients = [
```

7. Python Source Code (continued)

```
        {"name": "Anu", "consultation": 500, "diagnostic": 1200,
         "treatment": 2500, "paid": 4200},
        {"name": "Ravi", "consultation": 700, "diagnostic": 1800,
         "treatment": 4000, "paid": 4000},
        {"name": "Sara", "consultation": 300, "diagnostic": 500,
         "treatment": 1200, "paid": 0}
    ]

    print("=" * 76)
    print("             HOSPITAL BILLING AND PATIENT CLASSIFICATION")
    print("=" * 76)
    print(f"{'Patient':<10}{'Subtotal':>12}{'Concession':>13}"
          f"{'Net Bill':>12}{'Paid':>10}{'Balance':>11}  Status")
    print("-" * 76)

    for patient in patients:
        try:
            result = calculate_bill(patient)
            print(f"{result['name']:<10}"
                  f"{result['subtotal']:>12.2f}"
                  f"{result['concession']:>13.2f}"
                  f"{result['net_bill']:>12.2f}"
                  f"{result['paid']:>10.2f}"
                  f"{result['balance']:>11.2f}  {result['status']}")
        except ValueError as error:
            print(f"Error for {patient['name']}: {error}")

    print("-" * 76)
    print("Policy: 10% concession when subtotal is Rs. 5000 or more.")
    print("Classification: Paid in Full, Partially Paid, or Unpaid.")
    print("Billing analysis completed successfully.")

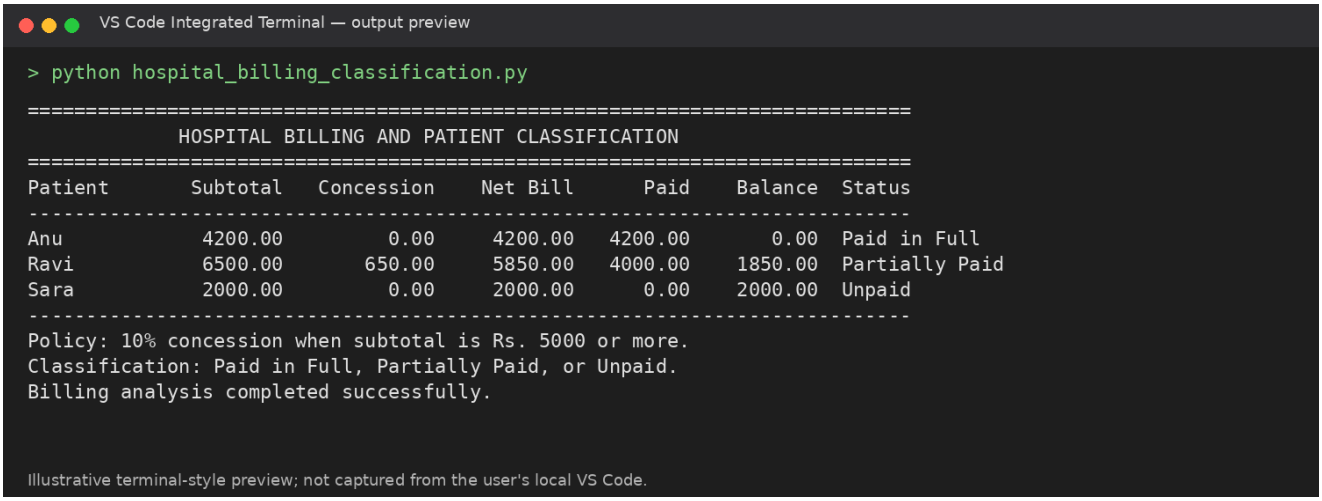

if __name__ == "__main__":
    main()
```

8. Sample Test Cases

Test case	Input / condition	Expected result
1. Normal bill	Subtotal ■4,200; paid ■4,200	No concession; balance ■0; Paid in Full
2. Concession	Subtotal ■6,500; paid ■4,000	10% concession ■650; net bill ■5,850; balance ■1,850; Partially Paid
3. Unpaid bill	Subtotal ■2,000; paid ■0	No concession; balance ■2,000; Unpaid
4. Negative charge	Any charge below zero	Show validation error
5. Excess payment	Paid amount greater than net bill	Show payment validation error

9. Output Screenshot

The image below is a terminal-style preview of the program output. It is not a direct screenshot captured from the user's local VS Code application.



To take a genuine screenshot in VS Code: open the .py file, select Terminal → New Terminal, run **python hospital_billing_classification.py**, then take a screenshot of the editor and terminal on your computer.

10. Conclusion

The program calculates hospital charges, applies a conditional concession, checks billing constraints, calculates the balance, and classifies patients based on payment status. The concession rule is an example assumption and should be changed if a different payment policy is supplied.